

Aquabolt-XL HBM2-PIM, LPDDR5-PIM With In-Memory Processing, and AXDIMM With Acceleration Buffer

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Processing-in-memory (PIM) has been proposed to improve the performance of bandwidth-intensive workloads as well as save energy due to reduced compute-memory data movement. To realize PIM, programmable computing units were integrated with memory cores on an HBM2 device to enable parallel processing and minimize data movement. A graphics processing unit (GPU) system equipped with Samsung Aquabolt-XL HBM2-PIM devices improved microkernel general matrix-vector multiplication and speech recognition applications by $8.9\times$ and $3.5\times$, respectively, and reduced energy consumption by over 60%. In a Xilinx AlveoU280 system, microkernel GEMV and ADD workload performances improved by $2.8\times$, and long short-term memory workload improved by $2.54\times$. Simulations show that a performance gain of over $2.3\times$ may be attained in a system with LP5-PIM for certain transformer-based speech recognition with an energy reduction of 86%. In addition, AXDIMM, a DIMM-level PIM with acceleration buffers, exhibits an 80% performance improvement and a 42.6% energy savings over a regular RDIMM system.

In the past decade, researchers of industries and academia have developed numerous artificial intelligence/machine learning (AI/ML) hardware accelerator systems to improve the accuracy and capability of AI/ML models. The success of these AI/ML models in turn drove the explosion in AI-optimized applications, and the demand for more accurate and more capable AI-optimized applications, in turn, fed the demand for more performance from the hardware acceleration systems. Depending on the application, the performance of hardware acceleration systems may be broadly classified as logic bound or memory bound, and some applications

may have different portions of the application runtime being logic bound or memory bound, respectively.¹ Consequently, high-bandwidth HBM2 and GDDR6 dynamic random access memory (DRAM) devices, and bandwidth power-efficient LPDDR5 and LPDDR5X DRAM devices have become first-order factors in determining the performance of memory-bound applications, such as speech recognition and machine translation. In this work, we discuss the implementation of processing-in-memory (PIM) technology that improves the performance of memory-bound AI/ML applications. We believe that future applications will require even higher memory bandwidth and lower energy and that PIM-enable memory can be used to target new solutions and provide both higher memory bandwidth and improved energy efficiency, as shown in Figure 1.² A simple solution to improve the performance of the memory-bound application is to provide improved memory bandwidth.

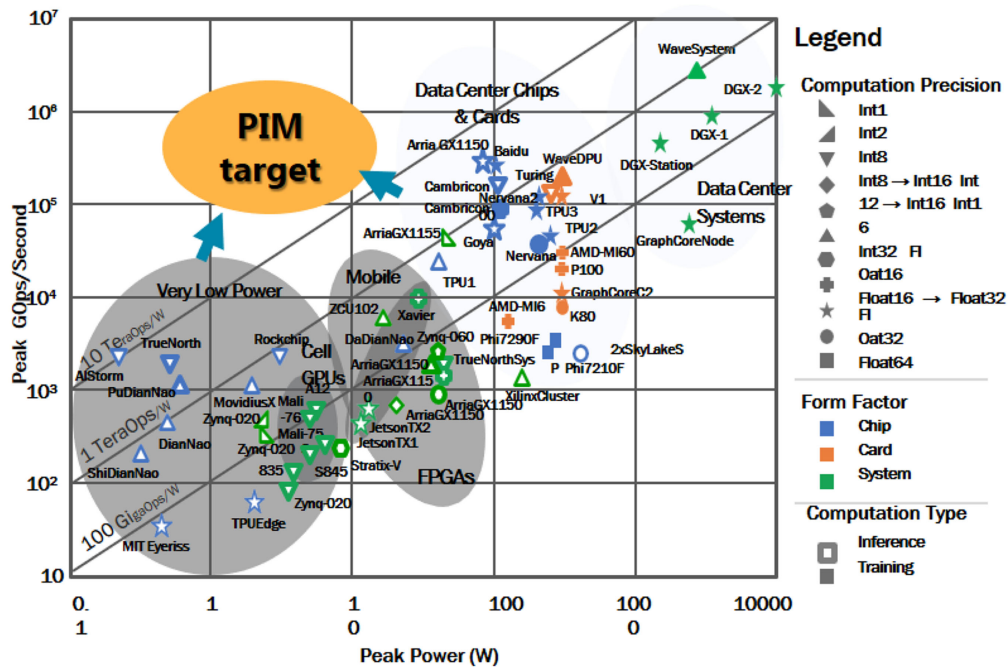


FIGURE 1. PIM target position on power for performance.

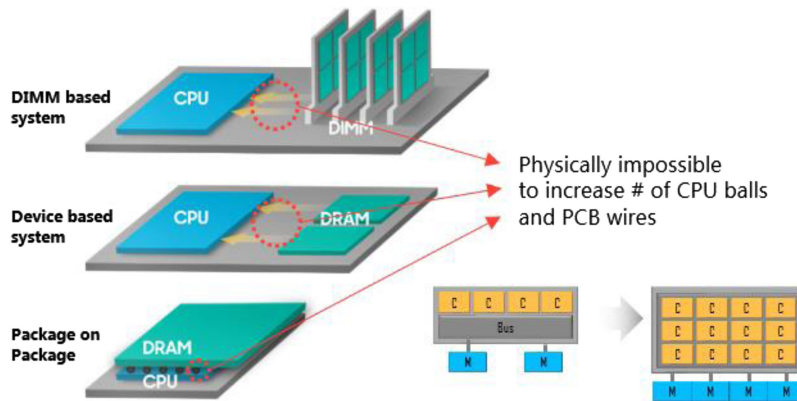


FIGURE 2. Increase existing memory bandwidth.

However, due to the physical constraints of chip packages and system boards, it is very expensive to further increase the bandwidth of central processing unit (CPU) to off-chip memory, as shown in Figure 2.^{3,4} The industry found that the overall performance of the system was limited by the PIN bandwidth of the off-chip, and to attain cost-effective bandwidth improvement and energy efficiency, the industry has begun to rethink and reinvestigate PIM architecture, as shown in Figure 3.⁵ In spite of recent advances, prior to our work, no major memory vendor had developed PIM memory, even as a proof-of-concept silicon. The reason for the lack of prior efforts on PIM is that previous PIM architectures often

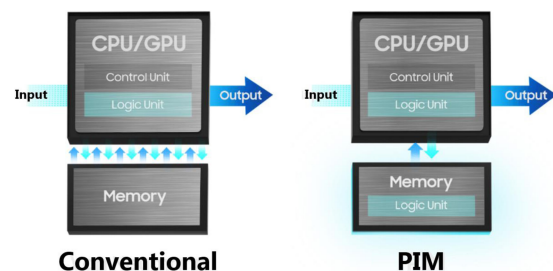


FIGURE 3. PIM concept.

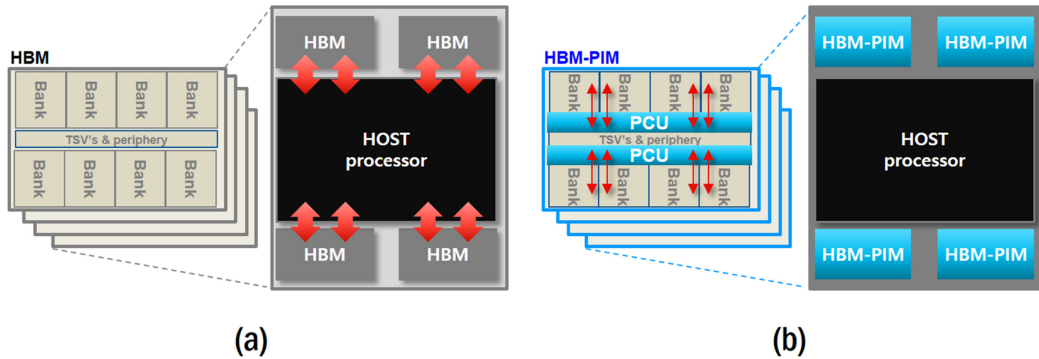


FIGURE 4. (a) Conventional HBM2. (b) HBM-PIM architecture.

required changes to the host processor and/or application code—conditions that could not be controlled by a memory vendor. Conventional memory is optimized for simple data reads and writes and is always coherent with the host's cache. SPEC of traditional memory did not consider the possibility of PIM at all, and the host or the application code had to be changed to optimize performance. Our approach differs from previous efforts by setting a bounding condition that a PIM architecture should improve performance and increase energy efficiency without requiring major changes to the host processor or application code.

A SIMPLE SOLUTION TO IMPROVE THE PERFORMANCE OF THE MEMORY-BOUND APPLICATION IS TO PROVIDE IMPROVED MEMORY BANDWIDTH.

PIM IMPLEMENTATION

PIM and Memory Hierarchy

To test out the concept of PIM, the Samsung Aquabolt-XL device was developed from the baseline of the currently shipping Samsung Aquabolt device with the integration of in-memory computing resources. The PIM-enabled HBM device is codenamed Aquabolt-XL. Aquabolt-XL was developed with the goal of improving on-chip computing bandwidth without changing the core DRAM design. A cache-line width single-instruction–multiple-data (SIMD) programmable computing unit (PCU) is integrated at the input/output (I/O) boundary of each bank in Figure 4. This architecture enables consistency in the use of cache-line-sized data access granularity as data accesses from the host processor. This architecture thus solves many problems associated with previous PIM architectures, such as data layout, virtual-to-physical memory address translation, and memory

interleaving. The Samsung Aquabolt-XL was developed with the intent for it to be used as a drop-in replacement for commercial systems using the Samsung Aquabolt HBM2 device.⁶ In addition, the new PIM instructions were aliased to backward-compatible column commands so that the host memory controller interfaces and platforms do not have to be modified to make use of the Aquabolt-XL HBM2-PIM device. This architectural decision further reduced the barrier of entry and allowed existing HBM2 systems to make use of the Aquabolt-XL HBM2-PIM device without requiring hardware changes to the host systems. To ensure compatibility, Aquabolt-XL HBM2-PIM preserves the external interface of HBM2 and is fully pin-compatible with existing processors with the HBM2 interface, allowing customers to maintain their existing form-factor and utilize existing HBM2 test infrastructures. For normal memory operations, Aquabolt-XL achieves the same performance and consumes the same amount of power as a standard Aquabolt HBM2 device. The compute engines in the memory can also be controlled by software using extended DRAM commands from existing HBM2 controllers. Special circuitry that cannot be bypassed at the software level has also been designed and implemented in Aquabolt-XL to prevent unintended use of the compute engines. Software enablement is intended to be simple, with Samsung providing PIM device drivers, APIs (TensorFlow and PyTorch), and PIM libraries for enabling PIM. In addition to layer mapping, proper placement of data in memory is also necessary to fully exploit the bank-level parallelism. The PCU is integrated with the memory core within Aquabolt-XL to enable parallel processing and minimize data movement. Modern parallel processors, such as GPGPUs or AI-centric processors are designed to accelerate high-density computing. These processors provide extremely high performance at the compute-bound layer but consume significant time and energy to process memory-bound operations with low arithmetic intensity. On the other hand, our PIM architecture can perform well in code regions with memory-bound operations, making PIM a natural complement to the host for

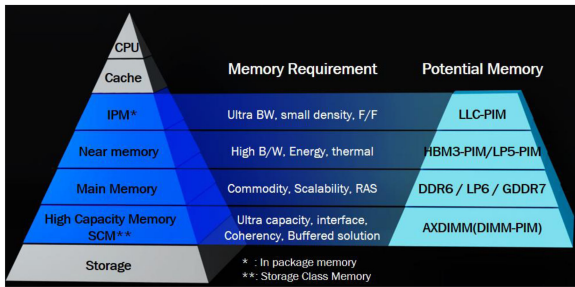
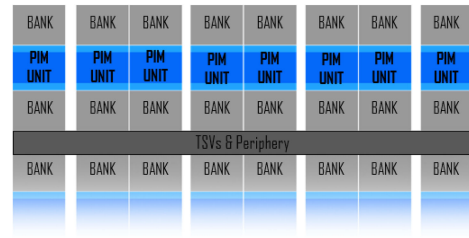


FIGURE 5. PIM device on potential memory hierarchy.

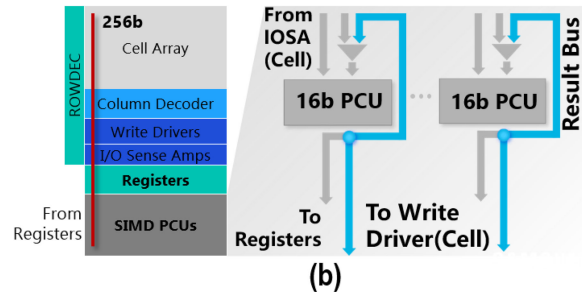
optimal system balance and performance/watt. When offloading memory-bound operations to the in-memory engine, this new architecture can deliver more than two-times system performance, while reducing energy consumption by more than 70% for memory-bound workloads, such as speech recognition, natural language translation, and recommendations. Many previous attempts have been made to divide the memory hierarchy and attain improved total cost of ownership (TCO) through the use of a hierarchical memory system. However, the presence of multiple tiers of memory without integrated computing within the memory hierarchy means that data movement through the heterogeneous memory stack will simply make the performance bottleneck imposed by the conventional architecture worse. To solve the fundamental problem of memory hierarchies creating new performance bottlenecks at the boundary of each layer of memory, PIM for various memory hierarchies is proposed, as shown in Figure 5.⁷

Aquabolt-XL HBM-PIM Architecture

In Samsung's first implementation of PIM, Aquabolt-XL, SIMD floating-point units have been placed at the bank-I/O-boundaries, near I/O-sense-amplifier arrays in Figure 6(a) to enable access to multiple banks/PCUs in a lockstep manner.⁸ Each PCU is connected to I/O-sense-amplifier and write-drivers, and interfaces with register files in Figure 6(b). However, we preserved DRAM bank control signals and corresponding logic circuits for standard HBM data accesses to the DRAM arrays. Conventional DRAM accesses the banks in parallel but serializes the bank data and loses vast internal bandwidth, whereas PIM makes high internal bandwidth available for computing across the bank. Also, since PIM computes for each column of all banks in parallel, the maximum speedup is proportional to the number of banks. Figure 7 introduces the Aquabolt-XL microarchitecture. Each PIM unit consists of three major components: an FP16 SIMD PCU array, register files, and a PIM unit controller. SIMD PCU array performs sixteen FP16 operations concurrently. The data used in the operations can be fetched from DRAM



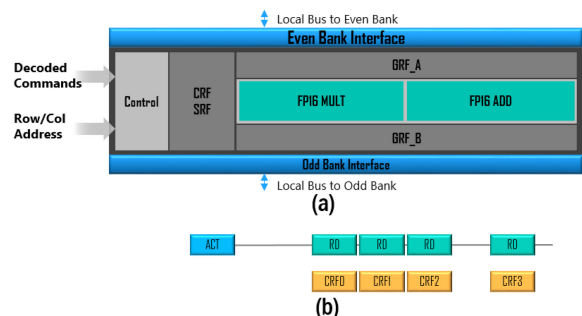
(a)



(b)

FIGURE 6. (a) HBM core die. (b) PIM UNIT on bank array.

bank, general register files, or scalar register files. The instructions for PIM operations can be stored in the command register files (CRFs), and a PIM unit controller controls all operations of the PIM unit such as fetching, decoding, controlling pipeline signals, and forwarding. In PIM mode, the PIM unit controller manages external data commands, such as read or write, increase CRF program counter and read or write data at the same time. Aquabolt-XL implements 32-bit RISC-style PIM instructions. In Table 1, four arithmetic, two data transfer and three control flow instructions are currently defined and supported. This PIM instruction set is a first implementation effort, and we are inviting academics, memory vendors, system partners, system customers, and application programmers to join us to co-define a more optimal PIM instruction set for a future, standardized version of PIM microarchitecture. We envision that the microarchitecture discussion would not be limited to the instruction



(b)

FIGURE 7. (a) PIM microarchitecture. (b) PIM memory command.

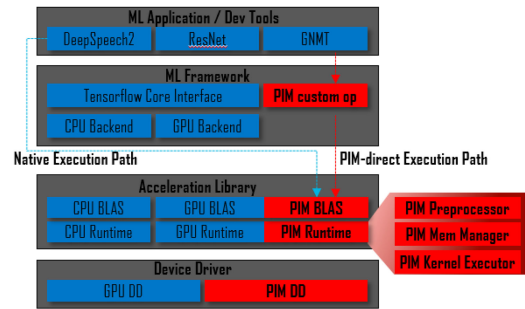
TABLE 1. Performance summary.

Type of DRAM	HBM2
Process	20-nm DRAM process
Memory density	6GB/Cube 4H(4Gb, PIM)+ 4H(8Gb)
Data rate per pin	2.4-Gbps per pin
Total bandwidth	307-GB/s per cube
# of channel	8 per cube
# of processing unit	128 per cube
Processing operation speed	300-MHz@measurement
Peak throughput	1.2TFLOPS per cube 300 MHz x 128 x 32 byte
Max operation precision	FP16

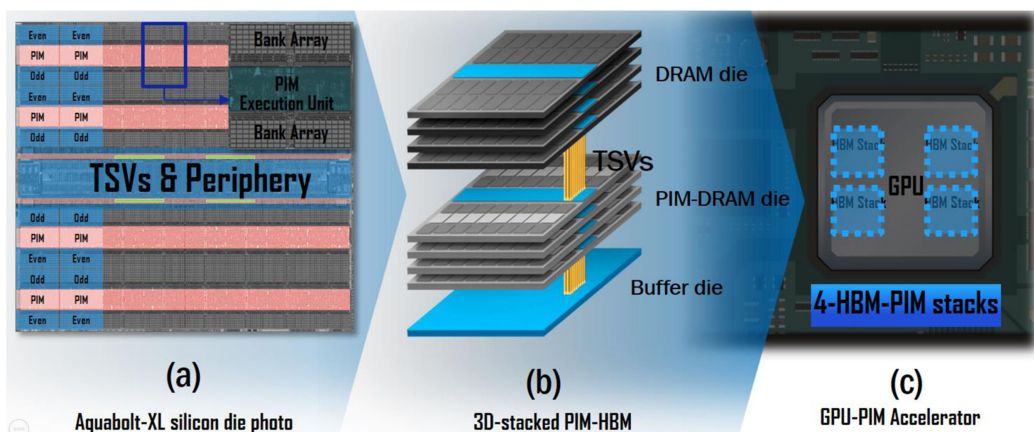
set, but will include all aspects of the PIM microarchitecture, such as supported data types, registers, and instruction formats.

PIM System Enabling and Test Results

A high-level desire in the definition of the Aquabolt-XL architecture and design is to make it possible to enjoy the benefits of PIM technology when running existing applications without SW modification. To achieve this desire, we examined and modified specific libraries and device drivers in the existing ML software stack. PIM-aware code represented by red blocks in Figure 8 is added to the existing ML software stack. In addition, to enable optimal performance, we also developed a direct execution path that allows the application programmer to access PIM instructions directly. The PIM-direct execution path is shown as a dotted red line in Figure 8. To implement PIM in silicon for Aquabolt-XL, we replaced the cell array area with the PIM execution block for

**FIGURE 8.** PIM software stack for GPU.

design convenience in Figure 9(a). In addition, to enable flexible testing of baseline HBM operations intermixed with HBM-PIM operations, we replace the bottom four DRAM dies of an eight-high HBM2 stack with PIM-enabled DRAM die while keeping the unmodified DRAM die in Figure 9(b). Thereafter, we assembled the eight-high HBM-PIM stack and coupled it to an unmodified, commercially available GPU/field-programmable gate array (FPGA) in Figure 9(c). Finally, we developed the supporting software stack to run existing applications without changing the source code (Figure 9). We evaluated the performance of the HBM-based PIM system using both microbenchmarks and end-to-end applications. We used matrix-vector multiplication and elementwise added microbenchmarks, Deep Speech 2 (DS2), RNN-T, and G-N-M-T for natural language processing application, and AlexNet and ResNet for computer vision applications. In Figure 10, the blue bars represent the performance gain on each benchmark from the PIM system compared to the baseline. To summarize the key results in the single batch case, the PIM system provides up to $8.9\times$ higher performance than the baseline, because the baseline

**FIGURE 9.** (a) Aquabolt-XL silicon dies photo. (b) 3-D-stacked HBM2-PIM. (c) GPU-PIM SiP package on accelerator.

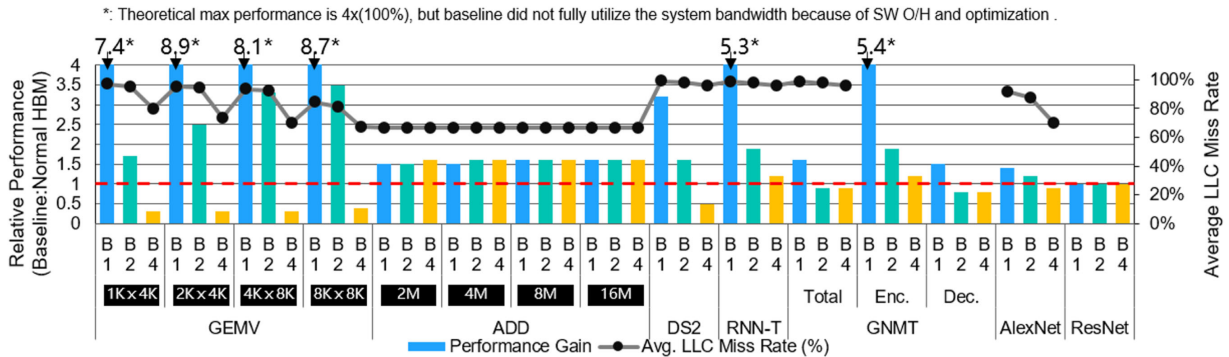


FIGURE 10. PIM performance comparison.

did not take full advantage of the system bandwidth. It shows high-performance improvement in GEMV microbenchmark, and in NLP end-to-end applications where GEMV is mainly used. For elementwise operations, such as ADD, increasing batch size, does not increase the data reusability, so the performance gain is the same as all batch sizes.⁹ We also measured the power consumption of the fabricated Aquabolt-XL HBM2-PIM in Figure 11 and found that Aquabolt-XL consumes only 5.4% higher power compared to the unmodified Aquabolt HBM2 device. During PIM operations, multiple PIM execution units operate concurrently, so the power consumption of DRAM internal components does increase, presented by gray and blue color bars in Figure 11, which is a hotspot. However, the power consumption of the internal global I/O bus, represented by green color bars, considerably decreases because of reduced data movement on the global bus. If the distance between the DRAM cell and the PIM unit increases as computed in the buffer die, the hotspot area increases, and the maximum internal memory bandwidth cannot be used. The power consumption of the buffer die and I/O PHYs, which is the orange bar, also decreases due to the decreased activity in PIM mode, but much power is currently wasted because the

bottom buffer dies was not modified for PIM. In the case that the bottom buffer die is optimized for PIM operations, the power consumption of HBM-PIM can be reduced further. Since we had to design lower than the design thermal limit in order to use Aquabolt's hardware infra as it is, we could not realize improved parallelism, and we could not change the buffer die due to the limited design limit. Work on HBM-PIM has just begun and much room remains for improvements and optimizations. We are presently identifying and targeting more memory-bound applications as candidates for performance acceleration and energy efficiency improvement. We show an example of the power consumption profiles of the DS2 execution in Figure 12. Due to the reduced processor utilization in the PIM-enabled system, the average power is much reduced. In addition, HBM-PIM reduces the execution time of DS2, and net energy efficiency improved significantly as a result.

FPGA PIM System Enabling and Test Results

We extended our PIM evaluation by changing the host from a GPU to an FPGA. For comparison purposes, we created two separate Xilinx Alveo U280 projects—a baseline FPGA system with an HBM device that does

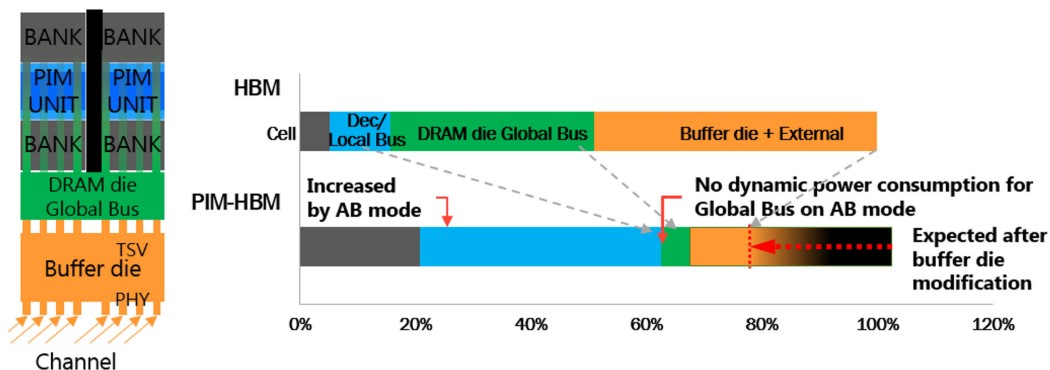


FIGURE 11. I/O power breakdown between HBM versus PIM-HBM.

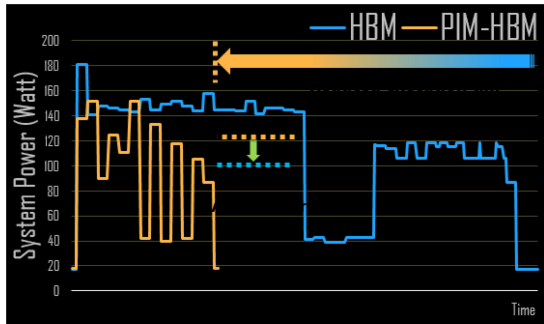


FIGURE 12. Average system power of DS2 over time.

not have PIM, and an FPGA system with the Aquabolt-XL HBM-PIM device. The baseline and PIM-enabled systems were designed to utilize the maximum HBM bandwidth. Figure 13 illustrates the high-level PIM evaluation architecture on the FPGA. We evaluated the performance of the FPGA-based PIM system using both microbenchmarks and end-to-end applications. We used matrix-vector multiplication, vector-vector Add, and GEMV matrix-based LSTM for microbenchmarks. To summarize key results in the microbenchmarks case, the PIM-enabled FPGA system provided $2.82\times$ higher performance than the baseline FPGA system in GEMV. In addition, ADD and LSTM microbenchmark results showed a performance gain of $2.85\times$ and $2.54\times$, respectively. We further evaluated the FPGA-based system enabled with an integrated HBM-PIM device using RNN-T end-to-end application. RNN-T application is mainly composed of LSTM models, as shown in Figure 14(a). The LSTM layer consumes around 90% of the running time, while the vector-matrix multiplication kernel uses around 80%. As a result, by optimizing for this bandwidth-bound region, PIM based FPGA system is able to deliver around $2.49\times$ the system performance while reducing energy consumption by 62% relative to the baseline system, as seen in Figure 14(b) and (c).

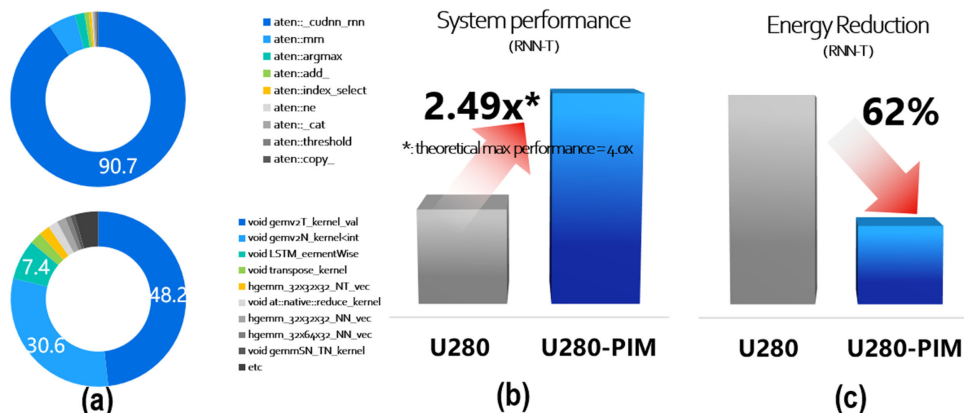


FIGURE 14. FPGA enabling (a) RNN-T workload analysis. (b) RNN-T performance. (c) RNN-T energy consumption.

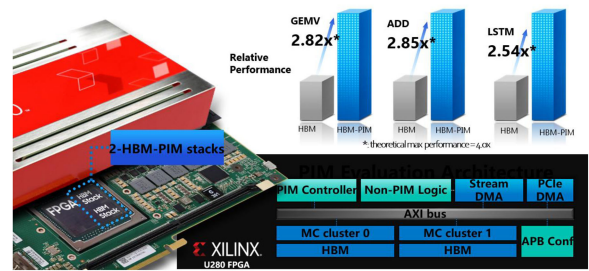


FIGURE 13. FPGA PIM HW/SW enabling and performance.

PIM EXPANSION OF AXDIMM AND LP5-PIM

AXDIMM, DIMM-Level PIM Introduction
PIM technology is extensible and can be applied at the module level as well as the component level. To investigate PIM technology at the module level, we implemented AXDIMM, the acceleration DIMM, as a concept testbed for module-level PIM. AXDIMM can maximize rank level parallelism by using all rank activation to overcome the performance bottlenecks of existing DDR channel topology, as seen in Figure 15. In addition to providing and exploiting rank-level parallelism to improve system performance and alleviate CPU-memory performance bottleneck, AXDIMM also provides specific acceleration functions in the AXDIMM buffer. Similar to HBM-PIM, AXDIMM can also reduce data movement energy by utilizing in-DIMM data processing unit and localizing data movement.¹⁰

AXDIMM Enabling and Test Results

As in the case of HBM-PIM, since the drop-in replacement is required without changing the existing ecosystem of CPU and application software, the AXDIMM support software stack was developed by adding the AXDIMM API, library, and device driver,

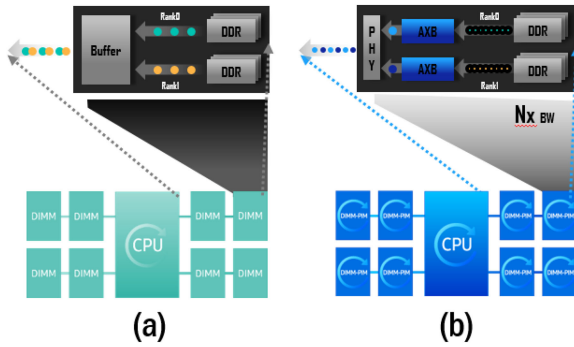


FIGURE 15. (a) Normal DIMM data movement. (b) AXDIMM data movement.

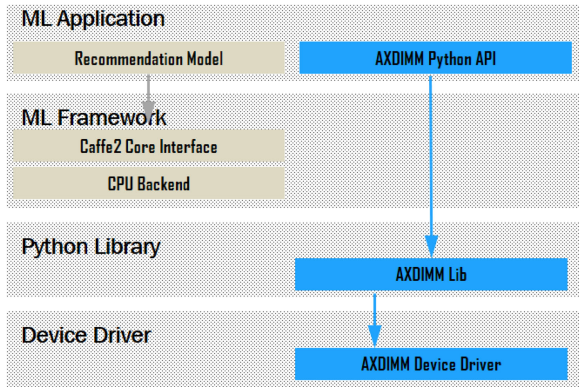


FIGURE 16. AXDIMM SW stack.

represented by blue boxes in Figure 16. Since AXDIMM exploits rank level parallelism, localized memory-compute bandwidth on an AXDIMM can scale linearly with the number of ranks on a given memory channel. We validated the speedup of embedding operations on AXDIMM from the expanded memory bandwidth. Our approach indicates that 80+% performance gain can

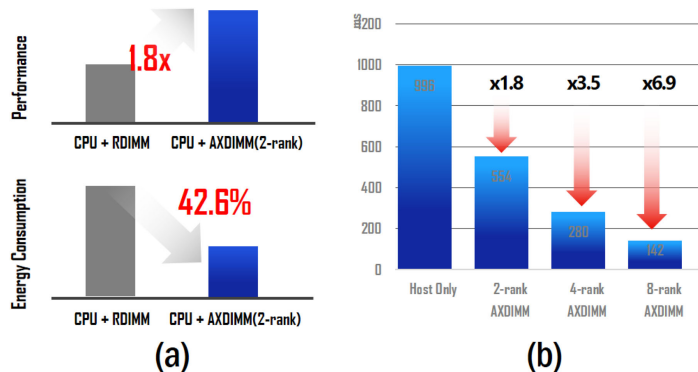
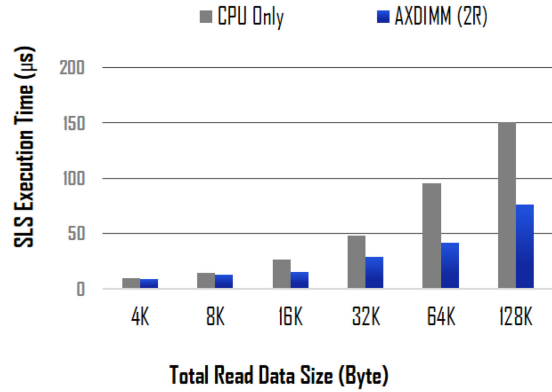
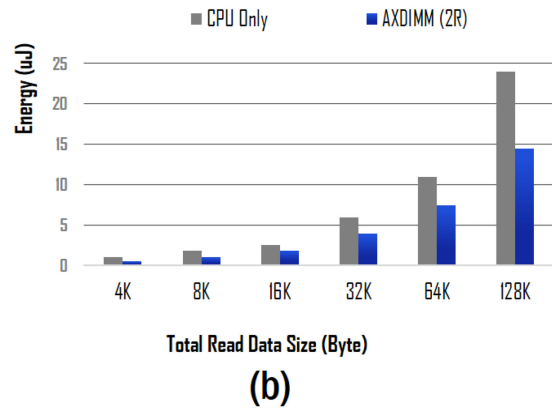


FIGURE 17. AXDIMM performance for extended rank.



(a)



(b)

FIGURE 18. (a) AXDIMM performance by data size. (b) AXDIMM energy by data size.

be realized in the two-rank configuration by utilizing the higher in-DIMM bandwidth with parallel operations across multiple ranks. In addition, 42.6% energy saving was observed for the two-rank configuration compared to the normal RDIMM system. We used SLS, a DLRM recommendation kernel to evaluate the near-memory processing AXDIMM. Figure 17 shows that using SLS, a two-rank AXDIMM achieves an 80% speedup over the baseline one-rank memory system; one-rank is a unit that can be activated at once. In addition, Figure 17 shows that performance scales nearly linearly with the increasing number of ranks, where 3.5 $\times$ speedup and 6.9 $\times$ speedup were observed for four-rank and eight-rank-configurations, respectively. A limitation when using AXDIMMs to expose a higher level of parallelism is the DIMM's form factor due to increased buffer size, which makes usage conditions difficult. Figure 18 shows the AXDIMM evaluation system and results. Figure 18 shows that the

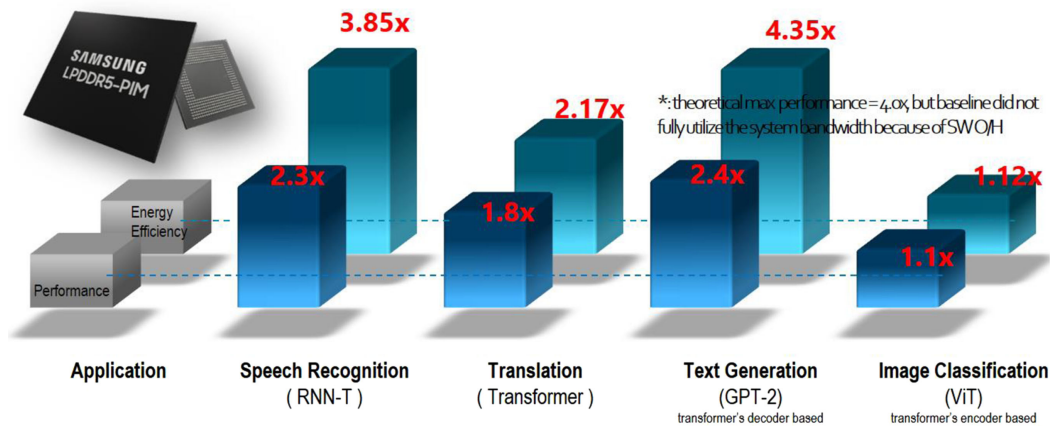


FIGURE 19. LPDDR5-PIM performance and energy.

memory footprint size of the SLS operator was increased incrementally from 4,000 to 128,000 for the 2-rank AXDIMM system. The AXDIMM system achieves plus 70%-to-80% latency speed-up across the tested memory footprint sizes by utilizing the doubled internal bandwidth of two-ranks on AXDIMM. The AXDIMM system also demonstrates significant memory energy saving by reducing the data movement between the host and AXDIMM, performing the embedding pooling in the memory system itself.

LPDDR5-PIM System Performance

As another example of the extension of PIM technology, performance of a simulated LPDDR5-PIM system as compared to a baseline system is shown in Figure 19. Figure 19 illustrates that PIM technology can be broadly applied to different classes of memory devices. In the present example, PIM technology is applied to LPDDR5 for low-power applications. Performance and energy efficiency were evaluated based on system level simulations. For comparison purposes, we assumed an NPU with 8.2 TFLOPS and 2.5 TFLOPS/W, 1-MB SRAM, and a one channel 12.8-GB/s LPDDR5X-6400 memory system. In the comparison system with PIM, RNN-T speech recognition model, transformer-based translation, and GPT-2 text generation showed performance gain of 2.3 $\times$, 1.8 $\times$, and 2.4 $\times$, respectively, and energy efficiency improvement compared to baseline system without PIM was 3.9 $\times$, 2.2 $\times$, and 4.4 $\times$ better, respectively. Compared to GPT-2, ViT showed performance and energy gain of only about 10% because ViT was composed mainly of compute-bound transformer encoder. However, many computer vision models, such as Image GPT and IPT, are instead based on transformer decoder, like GPT-2. We, therefore, believe that vision models may also be memory bound and can be accelerated with PIM technology.

CONCLUSION AND BEYOND

It should be mentioned why we need to reconsider this old PIM concept now. The industry already knows that process scaling is difficult and system power has a lot of influence on TCO. In addition, the heterogeneous programming model built by the GPU accelerator goes well with PIM technology, and various memory architectures using this model and various memory interfaces such as CXL are proposed. However, there were still some problems to be solved, and PIM presented some solutions that could be overcome through this article. This technology implemented more than twice the performance without changing the application code and focused only on basic functions that are faithful to the memory role, not very domain-specific functions, for fast-changing algorithms. In addition, the open source of SW will be promoted so that developers can easily use it to minimize the cost of paying for offload in the real world. Samsung's HBM-PIM device, codenamed Aquabolt-XL, was fabricated using a 20-nm DRAM process. To validate the performance of the fabricated Aquabolt-XL device, we constructed comparison systems using commercial off-the-shelf GPU and FPGA systems. We additionally provided optimized software stacks so that user code can be tested on Aquabolt-XL-enabled systems without rewriting the application code. Measured results indicate that the fabricated Aquabolt-XL device achieved 2.4-Gbps per pin at 1.2 V without increasing power consumption and that the integrated compute block, the PCU, was well integrated and worked well at 300 MHz. In addition, an emulation environment based on the application model was developed to confirm system-level performance and energy efficiency improvements by Aquabolt-XL-enabled systems. The estimated results appear to be promising as it shows a 2.7 $\times$ improvement in performance with 87% system power reduction. We are

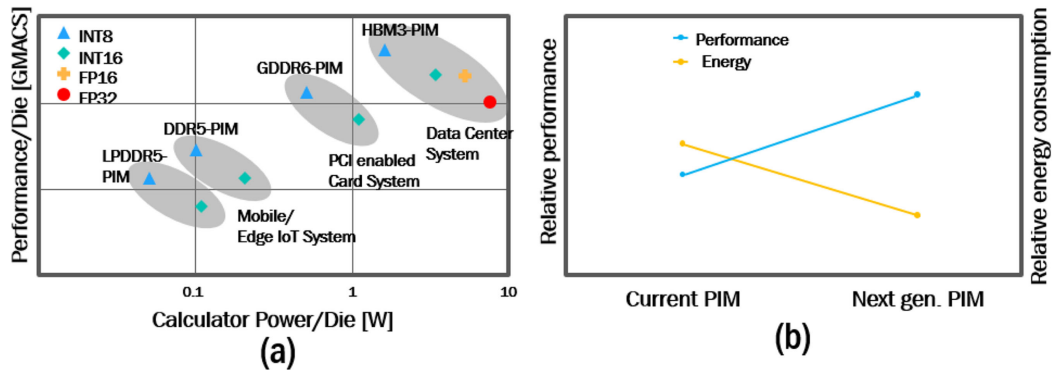


FIGURE 20. (a) Prospective PIM-supported data format. (b) Performance and energy of next PIM.

presently inviting academics, memory vendors, system partners, system customers, and application programmers to join us to co-define a new, more optimized PIM microarchitecture. Our plan is to investigate application areas and develop different sets of in-memory and near-memory processing frameworks for different classes of memory devices that may be standardized and commercialized for use. Figure 20 shows that LPDDR5-PIM may accelerate INT8 and INT16 data types for mobile/edge inference applications, and HBM3-PIM may accelerate all data types for high-performance AI applications. Figure 20 also shows that other memory devices with different classes of data type support may also be considered as customers and users identify new use cases and acceleration opportunities.

WE ARE PRESENTLY INVITING
ACADEMICS, MEMORY VENDORS,
SYSTEM PARTNERS, SYSTEM
CUSTOMERS, AND APPLICATION
PROGRAMMERS TO JOIN US TO
CO-DEFINE A NEW, MORE OPTIMIZED
PIM MICROARCHITECTURE.

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